EEG-Based Error Related Potentials

Sophia Kyemba

Alexander Steffey

Will Mulhern

Brody Greene

Ali Burkemper

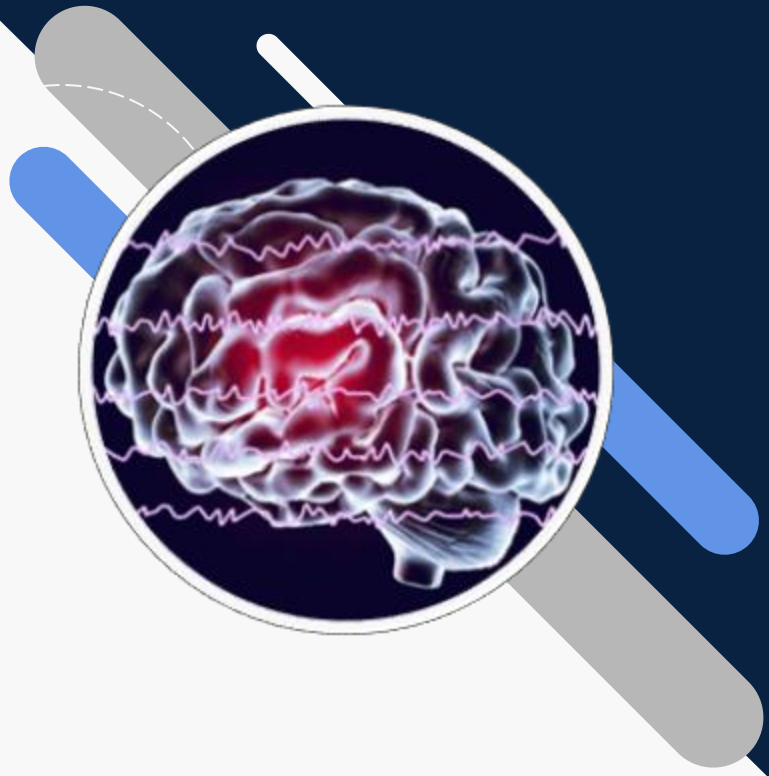
Project Site: https://aliburkemper12.github.io/Capstone-Project-Site/#/

GitHub: https://github.com/wrmulhern/CS495-EEG-Based-Error-Potentials

Project Focus

Creating a Low-Cost ErrP Validation Framework

- Visualize, record, & process EEG ErrP's in one place
- Easy and quick to use
- Native application
- Low number of dependencies
- Compatible with OpenBCI Ganglion
- Reduce reliance on MATLAB/MNE



Prioritized Sprint Goals

1. Test with Open BCI Ganglion hardware
2. Integrate flanker task
3. Code cleanup
4. Documentation
5. Unit testing

Sprint Achievements & Sponsor Feedback

Frontend

- Added GUI support for connection to BCI hardware and for running a flanker task

Backend

- Improved repo organization
- Found and fixed bugs
- Wrote unit tests

Feedback

- **Very positive -** Dr. Crawford was pleased with the progress made during the semester.
- He also demo'ed our application to the OpenBCI developers during a conference

Backlogs

Progress:

- Ganglion Device
- Flanker Task
- Security and Testing
- Documentation

ID	Story	Estimation (hours)	Priority (1-3)	Sprint When Finished
23	Convert csv to .set	4	1	2
24	Pressing 'enter' now lets user reload	2	2	2
25	Create a log system so unless we run it with the system in a debug mode we do not output info to tern	4	2	2
26	Get app running in cloud	7	1	2
28	Integrate with flanker task	10	1	3
29	Documentation per function (input, output, functionality)	3	1	3
30	Update ReadMe	2	1	3
31	Update help button to parse readme on github so if readme is updated the help is updated	2	2	3
32	Code cleanup	3	2	3
33	Test with OpenBCI Ganglion	6	1	3
34	Project site update with documentation	2	1	3
35	Unit tests	4	3	3
36	Icon once pyinstaller is working	2	3	3
37	Add flag info to ReadMe	3	2	3
38	Handle continuous (non-epoched) EEG data	12	1	3
39	Detect ErrPs in real-time during task execution	8	3	3
40	Add port description	1	1	3
41	User guide doc	2	1	3
42	Development guide doc (how to modify)	2	1	3
43	Install/Deployment guide doc	2	1	3
44	Test plan/cases doc	2	1	3
45	FAQs/Gotchas doc	2	1	3
46	Event dropdown (highlight/show different event epochs/time occured ?)	5	1	
47	Implement tools for ErrP detection	4	1	

Cyber Assessment

1. **Device Authenticity Verification:** confirms device is the proper ganglion board, re-checks board ID before streaming
2. **Secure Error Handling:** safe logging to not expose system details, clean error messages to not reveal sensitive information

Incomplete Work & Lessons Learned

Not Completed

1. Preprocessing and filtering newly captured EEG data
2. Deeper analysis of data using techniques like machine learning

Lessons

1. Extensive documentation
2. Preparing project for handoff
3. Integrating Ganglion Device

Accessibility & Inclusive Design

Traditional EEG tools require expensive licenses, programming knowledge, and complex setups, creating barriers for high school students, undergraduates, and new researchers.

Our accessible approach:

- **No-code GUI**: Drag-and-drop interface accessible to non-programmers
- **Free & Open Source**: No MATLAB license required
- **One-command install**: Standalone executable
- **Affordable hardware**: $625 Ganglion headset vs. $5000+ medical systems
- **Accessibility options**: Dark mode, keyboard shortcuts, clear error messages
- **Lightweight application**: 120 MB (OpenBCI GUI is ~250 MB)

Team Member Contributions

Sophia Kyemba

- Test development and documentation
- Ganglion hardware testing
- Added port descriptions
- Formatted backlogs
- Wrote FAQs doc

Alexander Steffey

- Added port and device security
- Test case development
- Test documentation
- Various GUI and bug fixes

Will Mulhern

- Worked on code-level documentation for various files
- Worked on bug fixes and misc updates
- Wrote user guide

Team Member Contributions

Brody Greene

- Worked on improving repo structure
- Worked on bug fixes and optimizations
- Wrote the Installation and Deployment Guide

Ali Burkemper

- Integrated Flanker task and EEG recording pipeline
- Created and implemented icon for executable
- Wrote the Development Guide
- Updated Project Site
- Bug fixes

Live Demo